SYLLABUS : UNIT & DIMENSIONS & MEASUREMENT ERROR AND EXPERIMENTS

1. The length of a rectangular plate is measured by a meter scale and is found to be 10.0 cm. Its width is measured by vernier callipers as 1.00 cm. The least count of the meter scale and vernier callipers are 0.1 cm and 0.01 cm respectively (Obviously). Maximum permissible error in area measurement is -

(A) + 0.2 cm^2 (B) + 0.1 cm^2 (C) + 0.3 cm^2 (D) Zero

2. For a cubical block, error in measurement of sides is + 1% and error in measurement of mass is + 2%, then maximum possible error in density is -

(A) 1% (B) 5% (C) 3% (D) 7%

3. To estimate 'g' (from $g = 4\pi^2 \frac{L}{T^2}$), error in measurement of L is + 2% and error in measurement of T is + 3%. The error in estimated 'g' will be -

(A) + 8% (B) + 6% (C) + 3% (D) + 5%

4. The least count of a stop watch is 0.2 second. The time of 20 oscillations of a pendulum is measured to be 25 seconds. The percentage error in the time period is

(A) 16% (B) 0.8 % (C) 1.8 % (D) 8 %

5. The dimensions of a rectangular block measured with a vernier callipers having least count of 0.1 mm is 5 mm × 10 mm × 5 mm. The maximum percentage error in measurement of volume of the block is

(A) 5 % (B) 10 % (C) 15 % (D) 20 %

6. An experiment measures quantities x, y, z and then t is calculated from the data as $t = \frac{xy^2}{z^3}$. If percentage errors in x, y and z are respectively 1%, 3%, 2%, then percentage error in t is :

(A) 10 % (B) 4 % (C) 7 % (D) 13 %

7. The external and internal diameters of a hollow cylinder are measured to be (4.23 ± 0.01) cm and (3.89 ± 0.01) cm. The thickness of the wall of the cylinder is

(A) (0.34 ± 0.02) cm (B) (0.17 ± 0.02) cm (C) (0.17 ± 0.01) cm (D) (0.34 ± 0.01) cm

8. The mass of a ball is 1.76 kg. The mass of 25 such balls is

(A) 0.44 × 10^3 kg (B) 44.0 kg (C) 44 kg (D) 44.00 kg

9. Two resistors R_1 (24 ± 0.5) Ω and R_2 (8 ± 0.3) Ω are joined in series. The equivalent resistance is

(A)32 ± 0.33 Ω (B) 32 ± 0.8 Ω (C) 32 ± 0.2 Ω (D) 32 ± 0.5 Ω

10. The pitch of a screw gauge is 0.5 mm and there are 100 divisions on it circular scale. The instrument reads +2 circular scale divisions when nothing is put in-between its jaws. In measuring the diameter of a wire, there are 8 divisions on the main scale and 83rd circular scale division coincides with the reference line. Then the diameter of the wire is

(A) 4.05 mm (B) 4.405 mm (C) 3.05 mm (D) 1.25 mm

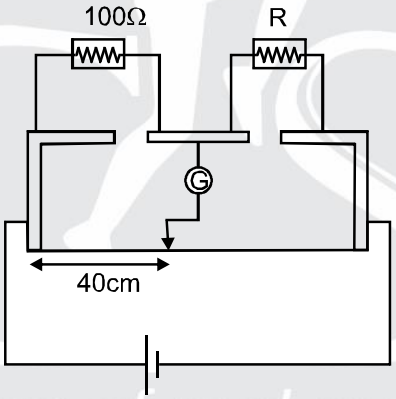
11. The pitch of a screw gauge is 1 mm and there are 50 divisions on its circular scale. When the two jaws of the screw gauge are in contact with each other, the zero of the circular scale lies 6 division above the line of graduation. When a wire is placed between the jaws, 3 main scale divisions are clearly visible while 31st division on the circular scale coincide with the reference line. The diameter of the wire is :

(A) 3.62 mm (B) 3.50 mm (C) 3.5 mm (D) 3.74 mm

12. The smallest division on the main scale of a vernier callipers is 1 mm, and 10 vernier divisions coincide with 9 main scale divisions. While measuring the diameter of a sphere, the zero mark of the vernier scale lies between 20 and 21 mm and the fifth division of the vernier scale coincide with a main scale division. Then diameter of the sphere is

(A) 20.5 mm (B) 21.5 mm (C) 21.50 mm (D) 20.50 mm

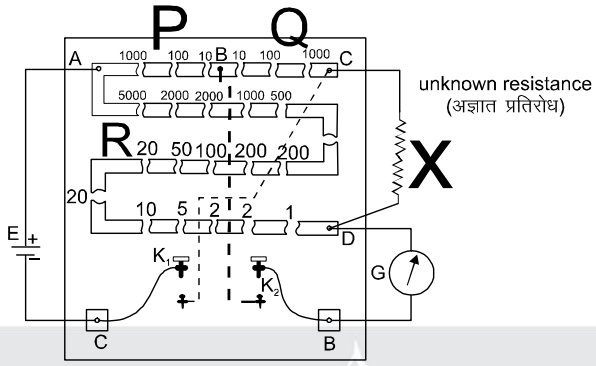
13. In a meter bridge circuit, the known resistance used in resistance box is 100 Ω (without any error, and the unknown resistor is put in right arm). The null point is found to be 40 cm from left end. If mm scale is used in the meter bridge then resistance of the unknown resistor is :



(A) $150\Omega \pm \frac{3}{8}\Omega$ (B) $150\Omega \pm \frac{3}{8}\Omega$ (C) $150\Omega \pm \frac{5}{8}\Omega$ (D) $150\Omega \pm \frac{7}{16}\Omega$

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14. In the post office box circuit, 10 Ω plug is taken out in arm AB and 100 Ω plug is taken out in arm BC. If the unknown resistor is kept in melting ice chamber, 600 Ω resistance is required in arm AD for zero deflection in galvanometer. Now if the unknown resistor is kept at 100° C (steam chamber), 630 Ω resistance is required in arm AD for zero deflection. Temperature coefficient of resistance of the unknown wire is :



(A) 2.5×10^{-4} / C° (B) 5×10^{-4}/ C° (C) 7.5×10^{-4} / C° (D) 8×10^{-4} / C°

15. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are

(A) 5, 1, 2 (B) 5, 1, 5 (C) 5, 5, 2 (D) 4, 4, 2

16. A screw gauge gives the following reading when used to measure the diameter of a wire .

Main scale reading : 0 mm

Circular scale reading : 52 division

Given that 1 mm on main scale corresponds to 100 divisions of the circular scale.

The diameter of wire from the above data is :

(A) 0.52 cm (B) 0.052 cm (C) 0.026 cm (D) 0.005 cm

17. If 400 Ω of resistance is made by adding four 100 Ω resistances of tolerance 5%, then the tolerance of the combination is :

(A) 5% (B) 10 % (C) 15 % (D) 20 %

18. The Current voltage relation of diode is given by $I = (e^{1000V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree Kelvin. If a student makes an error measuring ± 0.01V while measuring the current of 5 mA at 300K, what will be the error in the value of current in mA ?

(A) 0.2 mA (B) 0.02 mA (C) 0.5 mA (D) 0.05 mA

19. A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it?

(A) A meter scale.

(B) A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.

(C) A screw gausge having 100 divisions in the circular scale and pitch as 1 mm.

(D) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.

20. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90s using a wrist watch of 1s resolution. The accuracy in the determination of g is :

(A) 2% (B) 3% (C) 1% (D) 5%

21. A student measures the time period of 100 oscillations of a simple pendulum four times. That data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be :

(A) 92 ± 5.0 s (B) 92 ± 1.8 s (C) 92 ± 3 s (D) 92 ± 2 s

22. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5mm and the 25th division coincides with the main scale line?

(A) 0.80 mm (B) 0.70mm (C) 0.50mm (D) 0.75mm

23. The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is :

(A) 4.5% (B) 6% (C) 2.5% (D) 3.5 %

24. The pitch and the number of divisions, on the circular scale, for a given screw gauge are 0.5 mm and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 divisions below the mean line. The readings of the main scale and the circular scale, for a thin sheet are 5.5 mm and 48 respectively, the thickness of this sheet is :

(A) 5.740 mm (B) 5.950 mm (C) 5.725 mm (D) 5.755 mm

25. The diameter and height of a cylinder are measured by a meter scale to be 12.6 ± 0.1 cm and 34.2 ± 0.1 cm, respectively. What will be the value of its volume in appropriate significant figure.

(A) 4264 ± 81 cm^3 (B) 4300 ± 80 cm^3 (C) 4260 ± 80 cm^3 (D) 4264.4 ± 81.0 cm^3

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ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(A)	**4.**	(B)	**5.**	(A)
6.	(D)	**7.**	(C)	**8.**	(B)	**9.**	(B)	**10.**	(B)
11.	(D)	**12.**	(A)	**13.**	(C)	**14.**	(B)	**15.**	(A)
16.	(B)	**17.**	(A)	**18.**	(A)	**19.**	(B)	**20.**	(B)
21.	(D)	**22.**	(A)	**23.**	(A)	**24.**	(D)	**25.**	(C)